01 · PROBLEM UNDERSTANDING

Real-Time Fraud Detection for Small Merchants

Small merchants using digital payments are increasingly targeted by fraud, often with little protection. FraudShield is a proposed real-time system to help detect and prevent payment fraud for small businesses.

Problem Statement ID
Omni_FinTech_3

Track
FinTech

Practical challenges merchants face today

- **Slow identification.** Unusual transactions are difficult for a small team to spot quickly, especially in real time.
- **No dedicated fraud team.** Most small merchants have nobody watching transactions full time.
- **Noisy rule-based alerts.** Static rule engines tend to generate excessive false alerts, which merchants learn to ignore.
- **No "why."** A binary fraud / not-fraud label gives no explanation the merchant can act on.
- **Financial impact.** Every hour of delayed detection is a direct, avoidable loss for a small business.

Who has this problem

- Small retail merchants
- Local businesses
- Small online sellers
- Micro & small enterprises

Why existing tools fall short

- Expensive for a small merchant's budget
- Complex to configure without a technical team
- Limited explainability behind each alert
- Not built around the merchant's own behavior

Why this matters now

- Digital payment adoption among small merchants keeps growing, and fraud exposure grows with it.
- Fraud tactics evolve quickly — static, one-size-fits-all rules fall behind fast.
- For a thin-margin small business, even one large fraudulent transaction can be a serious setback.

Small merchants need **affordable, real-time, explainable** fraud protection that helps them make the right decision before a suspicious transaction becomes a financial loss.

02 Proposed Solution

FraudShield

An explainable, real-time fraud intelligence platform built specifically for small merchants — it does not just flag a transaction, it tells the merchant why, and what to do next.

STEP 1 Transaction → STEP 2 Real-Time Analysis → STEP 3 Risk Scoring → STEP 4 Explainability → STEP 5 Merchant Action → STEP 6 Feedback

01 Detect

Analyze every transaction in real time using transaction-level and behavioral signals as it happens.

Velocity · Amount · Device / session · Timing

02 Explain

Surface the specific factors contributing to the risk score — not just a label.

91 / 100 HIGH RISK

SIMULATED EXAMPLE — NOT A MEASURED RESULT

- Amount well above merchant's normal range
- Unusual transaction time
- Multiple failed attempts beforehand
- New device / session

03 Act

Translate the score into one clear, recommended action for the merchant to take.

APPROVE · REVIEW / VERIFY · BLOCK

04 Learn

Merchant feedback continuously refines detection and each merchant's behavioral baseline.

"Fraud" · "Legitimate" · "Reviewed"

FraudShield combines behavioral anomaly detection, adaptive risk scoring, explainable alerts and merchant feedback into one lightweight fraud-prevention platform built for small businesses.

FraudShield surfaces risk and recommends action — it does not claim guaranteed fraud prevention or 100% detection accuracy.

03 System Architecture

Every transaction moves through validation, a dual detection layer, adaptive scoring and an explainability pass before the merchant sees a recommendation — with feedback flowing back in to keep the model current.

Digital Payment / Transaction Source
POS · online checkout · payment gateway

Real-Time Data Ingestion API
Streams each transaction as it occurs

Transaction Validation & Feature Engineering
Cleans, structures and enriches each event

A Rule-Based Detection Engine

- Velocity checks
- Amount thresholds
- Repeated failed attempts
- Unusual timing
- Device / session anomalies

B ML Fraud Detection Engine

- Behavioral anomaly detection
- Transaction classification
- Merchant-specific patterns

Risk Scoring Engine
Combines both engines into one 0–100 score

Explainable AI Layer
Attributes the score to specific risk factors

Decision Engine
Routes the transaction by risk tier

Low Risk	Medium Risk	High Risk
Approve	Review	Block / Verify

Merchant Dashboard + Real-Time Alerts
Where the merchant sees and acts on it

Feedback loop
Merchant marks alerts as fraud / legitimate / reviewed

Merchant Feedback → Feedback / Learning Layer → Model & Risk Profile Improvement

Platform foundations

Shared infrastructure supporting every layer above.

- PostgreSQL / Supabase database
- Authentication & access control
- Real-time API layer
- Monitoring & logging
- Secure, encrypted data handling

Why a two-engine approach

Rules catch known, well-understood fraud patterns instantly and transparently, with no training data required.

The ML layer catches subtler, evolving behavioral anomalies that fixed rules alone would miss.

04 Key Features

01 Real-Time Transaction Monitoring
Continuously watches transactions as they happen and flags suspicious activity with minimal delay.

02 Dynamic Fraud Risk Score
Produces one understandable 0–100 risk score per transaction, instead of a raw model output.

03 Explainable Fraud Alerts
Shows the contributing risk factors behind every alert, not just a "fraud detected" label.

04 Merchant Behavioral Baseline
Learns each merchant's normal transaction pattern and flags deviations from it specifically.

05 Smart Action Recommendations
Recommends Approve, Review, Verify or Block based on the transaction's current risk tier.

06 Feedback-Driven Improvement
Lets merchants mark alerts as legitimate or fraudulent so detection keeps improving over time.

07 Fraud Analytics Dashboard
One view of transaction volume, flagged transactions, fraud trends, risk distribution and daily / weekly patterns — with confirmed-fraud vs. false-positive feedback built in.

`Transaction volume` `Flagged transactions` `Risk distribution` `Weekly patterns`

What the merchant sees

05 Technology Stack & Implementation Plan

Technology stack

FRONTEND **React.js + Tailwind CSS** Merchant dashboard and alert interface.	BACKEND **FastAPI + Python** Transaction ingestion and scoring services.	MACHINE LEARNING **Scikit-learn / XGBoost** Anomaly detection + supervised classification.	DATABASE **PostgreSQL / Supabase** Transactions, risk profiles, feedback.
REAL-TIME LAYER **WebSockets** Live event processing and alert delivery.	VISUALIZATION **Recharts / Chart.js** Analytics dashboard and trend charts.	SECURITY **JWT + RBAC** Encrypted data, role-based access, secure APIs.	DEPLOYMENT **Vercel + Render / Supabase** Practical, low-ops cloud deployment.

Implementation roadmap

01 Foundation

What gets built first — the working core.

- Merchant authentication & profiles
- Transaction ingestion API
- PostgreSQL / Supabase database
- Initial rule-based detection
- Basic merchant dashboard

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02 Intelligence

What intelligence gets added on top.

- Transaction feature engineering
- Merchant behavioral baselines
- ML anomaly detection / classification
- Dynamic 0–100 risk scoring

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03 Prevention

How prevention actually works for the merchant.

- Explainable fraud alerts
- Real-time notifications
- Approve / Review / Verify / Block workflow
- Merchant feedback loop

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04 Validation & Deployment

How the system gets validated and shipped.

- Precision / recall evaluation
- False-positive analysis
- Security & performance testing
- Cloud deployment & scalability

Expected Impact Faster detection · Fewer losses · Explainable decisions · Affordable protection · Scalable intelligence

Can this actually be built? Every layer of the stack — React, FastAPI, Scikit-learn, PostgreSQL — is widely adopted and well documented, so a small team can realistically ship the Phase 1 MVP within the hackathon build window.

FraudShield is a proposed hackathon solution and MVP implementation plan. No claims of a trained model, measured accuracy or live deployment are made at this stage.